How-To Library

Informatica

Using the Random Sampling Option in Profiles

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Abstract

You can choose to run a profile on all the rows in a data object, first N number of rows, or a random sample of data in the data object. This article discusses the random sampling options in profiling and how to use the options based on your requirement.

Supported Versions

- Data Quality 10.1.1

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Overview

You can run a profile on all the rows in the data object to perform a complete data quality analysis of the data source. You can also run the profile on the first few rows, or run the profile on a random sample of rows based on your business requirement.

In Informatica Analyst and Informatica Developer, when you create or edit a column profile, you can select a sampling option in the profile wizard. After you choose to run the profile on a random sample of rows, the random sample algorithm chooses the rows at random in the data object to run the profile on. When you choose a random sampling option for column profiles, the Analyst tool and Developer tool performs drilldown on the staged data. This can impact the drill-down performance. When you choose a random sampling option for data domain discovery profiles, the Analyst tool and Developer tool performs drill down on live data.

A data analyst can use random sampling to predict the distribution of data in a source system or quickly find the data quality of a source. You can use random sampling when the data source has skewed distribution or asymmetrical distribution of data. You cannot use random sampling option for unstructured data sources.

Random Sampling Computation

The random sampling algorithm retrieves the total row count from the data source and computes the number of random sample rows. If the data source is a statistical database, such as Oracle, Microsoft SQL Server, or IBM DB2, then the algorithm gets the row count from the statistics API. For non-statistical databases, the Data Integration Service runs the ROW_COUNT mapping to retrieve the row count. The algorithm computes the number of random sample rows based on the random sampling option that you choose in the profile wizard.

If the data source is a relational data source, such as Oracle, Microsoft SQL Server, or IBM DB2 and supports random sampling of data, then the Data Integration Service pushes the SQL query to the database. For example, to select the random rows in the Customers table for profiling, a sample query is Select * from Customers SAMPLE (X) statement, where X is the approximate percentage of random rows. The query returns an approximate X percentage of rows on which the profile runs. For example, assume that the estimated source row count for the Customers table is 100 rows. The computed approximate percentage of random rows X is 0.35. The query might return 33 or 36 rows. This is because the query Select * from Customers SAMPLE (0.35) may or may not return 35 rows as a small difference in rows might exist between the query results and the computed percentage of random rows.

If the data source does not support the random sampling option, then the Data Integration Service runs a profiling custom transformation after it runs the source transformation. The profiling custom transformation passes the random sample rows downstream for column profile or data domain discovery profile computation.

You can choose one of the following types of random sampling options in the Analyst tool or the Developer tool:

Random sample (auto)

The random sample algorithm computes the percentage of sample rows based on the total row count in the data source. If the total row count is less than 1000, then the profile runs on 100% rows.

The following table shows the random sample algorithm computation based on the number of rows in the data object:

Data Source Row Count	Computed Percentage of Rows for Random Sampling
<1K	100%
1K to 10K	90%, 80%, 70% ...10%
10K to 100K	10%
100K to 1M	10%, 9%, 8% ... 1%
>1M	1%

Random sample

You can configure the number of random rows when you choose the Random sample option. The random sampling algorithm converts the absolute number of rows to percentage based on the source row count.

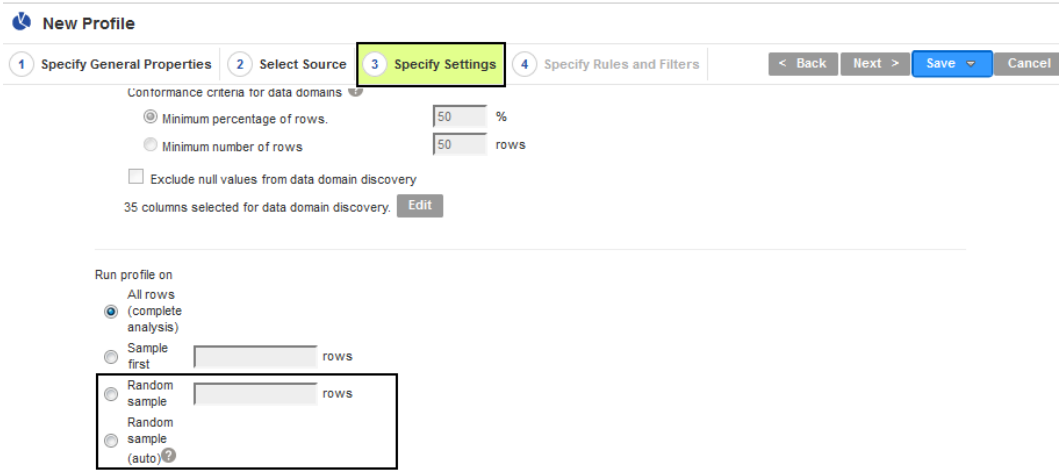
Using Random Sampling Option in Informatica Analyst

You can choose the random sampling option in the Analyst tool when you create or edit a column profile.

1. In the Analyst tool, click **New** > **Profile**.

   The profile wizard appears.
2. Choose **Single source** to create a column profile. Click **Next**.
3. In the **Specify General Properties** screen, enter a name for the profile, and choose a location to save the profile. Click **Next**.
4. In the **Select Source** screen, choose a data object. Click **Next**.
5. In the **Specify Settings** screen, choose the sampling option as **Random sample** or **Random sample (auto)** based on your requirements.

   The following image shows the sampling option in the **Specify Settings** screen in the Analyst tool:



6. Choose a drilldown option and the run-time environment for the column profile. Click **Next**.
7. In the **Specify Rules and Filters** screen, you can choose to add rules or filters.
8. Click **Save and Run** to run the profile, or click **Save and Finish** to save the profile.

Using Random Sampling in Informatica Developer

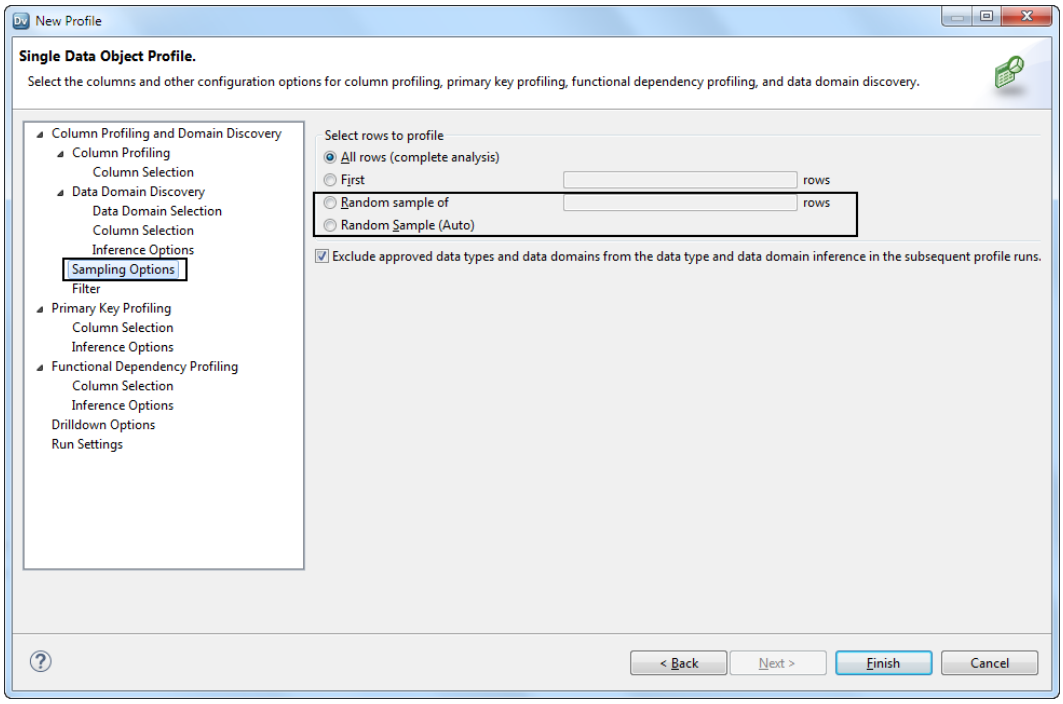
You can choose to run the column profile on a random sample of data in the data source in the Developer tool.

1. In the Developer tool, click **File** > **New** > **Profile**.

   The profile wizard appears.
2. In the profile wizard, choose **Profile** to create a column profile. Click **Next**.
3. In the **Configure general properties** screen, enter a name for the profile, and click **Add** to choose a data object. Select the **Run Profile on Finish** option to run the profile after you create the profile. Click **Next**.
4. Click **Sampling Options** in the **Column Profiling and Domain Discovery** section.

   The sampling options for the column profile appears.
5. Choose **Random Sample of** or **Random Sample (Auto)** option. If you choose the **Random Sample of** option, then choose the number of random rows to run the profile on.

   The following image shows the sampling options for a column profile in the Developer tool:



6. Choose a drilldown option and the run-time environment to run the profile. Click **Finish**.

   The profile runs on a random sample of data in the data source.

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